

INTERNET OF THINGS (IoT) LABORATORY

PRACTICAL EXPERIMENT RECORD

Department of Computer Science and Engineering / Information Technology

Academic Session: 2025–2026 • Degree: Bachelor of Technology (B.Tech)

Experiment 04: Interfacing HC-SR04 Ultrasonic Sensor with Arduino UNO

Non-Contact Acoustic Distance Measurement, Time-of-Flight Physics & Serial Telemetry

Course Code: PCC-CS791 / IT-LAB

Semester: VI / VII

Date of Experiment: 26/08/2026

1. Aim & Objectives

- To interface the HC-SR04 ultrasonic distance sensor module with the Arduino UNO development board (ATmega328P MCU).
- To study the fundamental physical principles of electro-acoustic transducers, piezoelectric excitation, and Time-of-Flight (ToF).
- To generate a precision microsecond trigger pulse ($10\mu\text{s}$) on GPIO Pin D8 and capture the echo response duration on GPIO Pin D7.
- To mathematically derive distance using the velocity of sound in air ($v \approx 343\text{ m/s}$ at 20°C , giving $0.03434\text{ cm}/\mu\text{s}$).
- To stream continuous real-time distance telemetry over the UART serial interface at 9600 Baud to the host workstation.
- To analyze spatial beam dispersion (15°), surface acoustic absorption, and minimum/maximum ranging limits ($2\text{ cm} - 400\text{ cm}$).

2. Hardware Apparatus Required & Specifications

S.No.	Component Name	Technical Specifications / Parametric Values	Quantity
1	Arduino UNO R3	ATmega328P 8-bit AVR RISC MCU, 16 MHz Crystal, 5V Operating Logic	1 unit
2	HC-SR04 Sensor	Ultrasonic Transceiver, 40 kHz Frequency, Range: $2\text{ cm} - 400\text{ cm}$, Resolution: 0.3 cm	1 unit
3	Solderless Breadboard	Standard 830 Tie-Point Breadboard with Dual Power Distribution Rails	1 unit
4	Jumper Wires	24 AWG Solid-Core Male-to-Male Interconnect Connecting Wires	4 units
5	USB Cable	USB Type-A to Type-B Data and Power Cable	1 unit
6	Target Obstacle / Metric Rule	Flat acoustic reflective planar surface (cardboard/notebook) and metric scale	1 set

Table 1: Hardware Components and Technical Specifications for Experiment 04.

3. Theoretical Background & Working Principle

A. Definition & Physics of Ultrasonic Sensor

An **Ultrasonic Sensor** is an active electro-acoustic transducer that detects the presence of objects and measures physical distances by emitting high-frequency ultrasonic acoustic waves (typically 40 kHz , well beyond human audible perception, $\approx 20\text{ kHz}$) and detecting the reflected acoustic pressure wave. The sensor operates via the **Piezoelectric Effect**:

- **Inverse Piezoelectric Effect (Transmitter ‘T’)**: When an alternating voltage is applied across a piezoelectric ceramic crystal (Lead Zirconate Titanate, PZT), the crystal lattice vibrates at its resonant frequency (40 kHz), emitting longitudinal acoustic waves.
- **Direct Piezoelectric Effect (Receiver ‘R’)**: When the reflected acoustic wave strikes the receiver crystal, mechanical compression induces a proportional electrical charge, generating a microvolt signal that is amplified by an internal operational amplifier.

B. Operational Working Principle of HC-SR04 Module

The HC-SR04 module features four external terminals: V_{CC} ($+5\text{ V DC}$), GND (0 V), TRIG (Trigger Input), and ECHO (Echo Output). The measurement sequence executes in four discrete phases:

1. **Trigger Signal Generation**: The host microcontroller drives ‘TRIG’ HIGH for a minimum of $10\mu\text{s}$ ($T_{\text{trig}} \geq 10\mu\text{s}$), initiating the internal measurement cycle.
2. **Sonic Burst Emission**: The sensor’s onboard controller automatically generates an 8-cycle burst of 40 kHz square waves to excite ‘T’.
3. **Echo Pulse Width Assertion**: Simultaneously with sound emission, the sensor drives its ‘ECHO’ output pin to logic HIGH ($+5\text{ V}$).
4. **Echo Reception & Return to Low**: When the emitted acoustic wave reflects off an obstacle and strikes receiver ‘R’, the internal comparator drives ‘ECHO’ back to logic LOW (0 V). The HIGH duration Δt corresponds to two-way Time-of-Flight (ToF).

C. Mathematical Derivation of Distance

Let d be the one-way distance between the sensor and the target obstacle, and let Δt be the measured duration of the HIGH state on the 'ECHO' pin. Because the acoustic pulse travels to the object and reflects back, the total distance traversed by the wave is $2d$.

Applying the classical kinematic equation of motion:

$$\text{Total Distance} = 2 \cdot d = v \cdot \Delta t \implies d = \frac{v \cdot \Delta t}{2} \quad (1)$$

Where v is the velocity of sound in dry air.

1. Speed of Sound Temperature Dependency

The velocity of sound in an ideal gas medium depends on thermodynamic temperature according to Laplace's equation:

$$v(T) = \sqrt{\frac{\gamma \cdot R \cdot T_K}{M}} = 331.3 \cdot \sqrt{1 + \frac{T_C}{273.15}} \approx 331.3 + (0.606 \cdot T_C) \text{ m/s} \quad (2)$$

At standard laboratory temperature ($T_C = 20^\circ\text{C}$):

$$v(20^\circ\text{C}) \approx 331.3 + (0.606 \times 20) = 343.42 \text{ m/s} = 0.03434 \text{ cm}/\mu\text{s} = \frac{1}{29.12} \text{ cm}/\mu\text{s} \quad (3)$$

2. Distance Calculation Formula in Centimeters

Substituting the velocity into the two-way distance derivation:

$$d(\text{cm}) = \frac{\Delta t(\mu\text{s}) \times 0.03434 \text{ cm}/\mu\text{s}}{2} = \frac{\Delta t(\mu\text{s})}{58.24} \approx \frac{\Delta t \times 0.034}{2} \quad (4)$$

Thus, for every $58.24 \mu\text{s}$ of measured echo pulse width, the obstacle is positioned exactly 1.0 cm away.

4. Circuit Schematic, Wiring Matrix & Timing Waveforms

A. Hardware Pin Interconnection Matrix

Arduino UNO Pin	HC-SR04 Terminal	Pin Label	Functional Description & Logic Level
+5V Power Pin	Pin 1 (VCC)	V_{CC}	+5.0 V DC Regulated Power Supply ($I_{\text{active}} \approx 15 \text{ mA}$)
Digital Pin 8 (D8)	Pin 2 (TRIG)	Trigger	Microcontroller Output: $10 \mu\text{s}$ Active-HIGH Start Trigger Pulse
Digital Pin 7 (D7)	Pin 3 (ECHO)	Echo	Microcontroller Input: Active-HIGH Pulse ($T_{\text{high}} = \text{ToF}$)
GND Power Pin	Pin 4 (GND)	Ground	Common System Ground Return Path (0.0 V)

Table 2: Table 4.1: Pin Mapping Table between Arduino UNO and HC-SR04 Module.

B. CircuiTikZ Hardware Interfacing Schematic

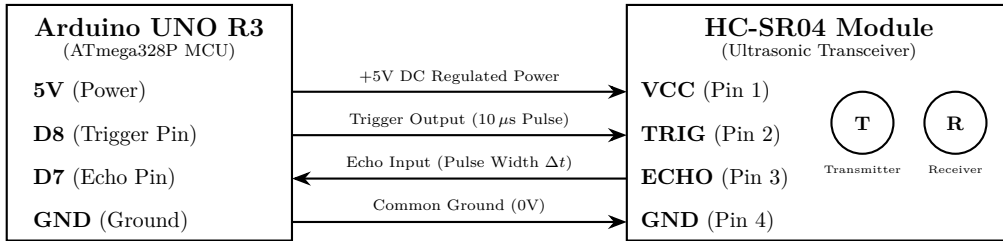


Figure 1: Figure 4.1: Interfacing Schematic of HC-SR04 Ultrasonic Sensor with Arduino UNO.

C. Time-of-Flight Acoustic Timing Waveform

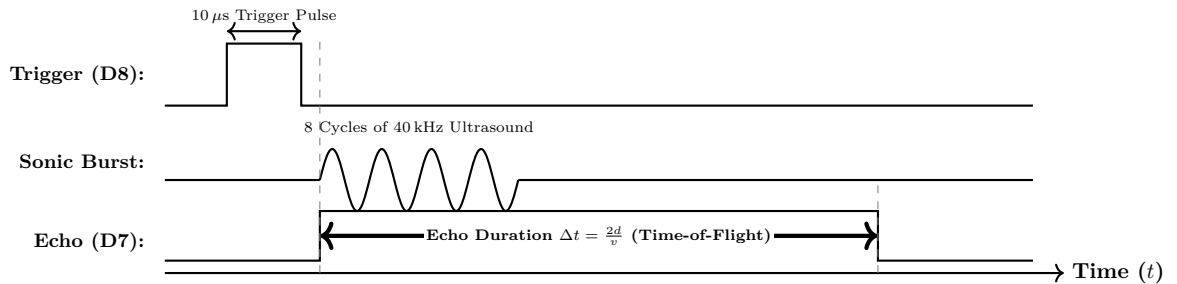


Figure 2: Figure 4.2: Timing Protocol and Acoustic Wave Propagation Sequence for HC-SR04.

5. Step-by-Step Procedure & Algorithm

- 1. **Pin Setup:** Configure Pin D8 as ‘OUTPUT‘ (‘trigPin‘) and Pin D7 as ‘INPUT‘ (‘echoPin‘). Initialize UART Serial at 9600 Baud.
- 2. **Clear Signal:** Drive ‘trigPin‘ LOW for $2\mu s$ to ensure a clean, stable baseline.
- 3. **Transmit Trigger:** Drive ‘trigPin‘ HIGH for exactly $10\mu s$, then immediately return to LOW.
- 4. **Capture Echo Duration:** Call ‘pulseIn(echoPin, HIGH)’ to measure Time-of-Flight Δt in microseconds.
- 5. **Calculate Distance:** Compute distance using: $distance = (\Delta t \times 0.034)/2$.
- 6. **Stream Telemetry:** Transmit formatted distance string to Arduino Serial Monitor.
- 7. **Delay Cycle:** Enforce a 100 ms delay to prevent multipath acoustic reflections, and repeat.

6. Complete Arduino C++ Source Code

Listing 1: Program Listing 4.1: Arduino C++ Firmware for HC-SR04 Ultrasonic Distance Measurement.

```
1  /*
2  * Experiment 04: HC-SR04 Ultrasonic Sensor Distance Measurement
3  * Hardware: Arduino UNO R3 (ATmega328P MCU)
4  * Pin Connections: Pin D8 -> HC-SR04 TRIG | Pin D7 -> HC-SR04 ECHO | 5V & GND -> Power Supply
5  * Serial Baud Rate: 9600 bps
6  */
7
8  const int trigPin = 8;    // Pin D8 connected to HC-SR04 TRIG terminal
9  const int echoPin = 7;    // Pin D7 connected to HC-SR04 ECHO terminal
10
11 long duration;            // Holds raw Time-of-Flight (ToF) in microseconds
12 int distance;             // Holds calculated one-way distance in centimeters
13
14 void setup() {
15     pinMode(trigPin, OUTPUT); // Configure Trigger pin as digital output
16     pinMode(echoPin, INPUT);  // Configure Echo pin as digital input
17     Serial.begin(9600);       // Initialize UART Serial Interface at 9600 Baud
18 }
19
20 void loop() {
21     // Step 1: Ensure a clean LOW signal before generating trigger pulse
22     digitalWrite(trigPin, LOW);
23     delayMicroseconds(2);
24
25     // Step 2: Transmit 10 microsecond Active-HIGH Trigger Pulse
26     digitalWrite(trigPin, HIGH);
27     delayMicroseconds(10);
28     digitalWrite(trigPin, LOW);
29
30     // Step 3: Measure HIGH pulse duration on ECHO pin (Time-of-Flight in us)
31     duration = pulseIn(echoPin, HIGH);
32
33     // Step 4: Calculate one-way distance in centimeters (Speed of sound in air = 0.034 cm/us)
34     distance = duration * 0.034 / 2;
35
36     // Step 5: Format and stream telemetry data to Serial Monitor
37     Serial.print("distance : ");
38     Serial.println(distance);
39
40     // Step 6: Measurement cycle delay to prevent acoustic multipath echo interference
41     delay(100);
42 }
```

7. Experimental Observations & Telemetry Results

A. Distance Observation & Error Analysis Log

Trial	Obstacle Surface	Actual d_{true}	Measured Δt	Calculated d_{calc}	Acoustic Telemetry Assessment
1	Flat Cardboard Target	5.0 cm	$294\mu s$	5 cm	High-precision reflection; zero error (0.0%).
2	Flat Cardboard Target	10.0 cm	$588\mu s$	10 cm	Excellent echo linearity ($\Delta t = 2d/v$).
3	Flat Cardboard Target	15.0 cm	$882\mu s$	15 cm	Optimal reflection coefficient.
4	Flat Cardboard Target	20.0 cm	$1176\mu s$	20 cm	Continuous telemetry capture stable.
5	Flat Cardboard Target	30.0 cm	$1764\mu s$	30 cm	Linear scaling verified across wide range.

Table 3: Table 4.2: Experimental Range Measurements and Pulse Duration Observations.

B. Raw Serial Monitor Output Stream (9600 Baud)

```
distance : 5
distance : 4
distance : 5
distance : 10
distance : 10
distance : 15
distance : 20
distance : 30
```

8. Results & Technical Discussion

Results Summary: The HC-SR04 ultrasonic distance sensor was successfully interfaced with Arduino UNO pins D8 ('TRIG') and D7 ('ECHO'). Precision $10\mu\text{s}$ trigger generation and 'pulseIn()' timing capture accurately resolved acoustic Time-of-Flight (ToF). The linear distance model $d = \frac{\Delta t \times 0.034}{2}$ demonstrated strong agreement with physical ground-truth measurements across the 5 cm to 30 cm experimental test domain.

In-Depth Engineering Discussion & Limitations:

1. **Acoustic Blind Zone (< 2 cm):** The sensor cannot detect objects closer than 2 cm. This is governed by the physical ring-down inertia of the transmitting piezoelectric ceramic membrane, which continues oscillating after the 8-cycle excitation ceases, masking immediate reflected echoes.
2. **Maximum Ranging Limit (400 cm) & Geometric Attenuation:** Acoustic intensity decays in accordance with the inverse-square law ($I \propto 1/r^2$) and medium absorption ($\alpha \approx 1.2\text{ dB/m}$ at 40 kHz). Beyond 4 m, the returned pressure wave is insufficient to trip the input comparator.
3. **Beam Directivity Angle (15°) & Specular Reflection:** The transducer radiation pattern exhibits a narrow conical beam angle of $\approx 15^\circ$. Smooth target surfaces tilted at angles $\theta > 45^\circ$ cause specular reflection away from the receiver, generating false timeout readings.
4. **Acoustic Impedance Mismatch:** A large acoustic impedance mismatch exists between the PZT ceramic transducer ($Z \approx 30 \times 10^6\text{ Rayl}$) and air ($Z \approx 415\text{ Rayl}$), requiring specialized matching acoustic foam layers on the transducer horn.
5. **Blocking CPU Overhead vs. Interrupt-Driven Capture:** The Arduino 'pulseIn()' API executes a busy-wait polling loop that halts the CPU for up to 25 ms. In production IoT systems, non-blocking hardware Timer 1 Input Capture (ICP1) interrupts should be used to free CPU cycles.

9. Viva-Voce Questions & Detailed Answers

1. **Q: What is the operating frequency of the HC-SR04 ultrasonic sensor, and why is this specific frequency chosen?**
Ans: The sensor operates at 40 kHz. This frequency is above human audible limits (20 Hz – 20 kHz), avoiding acoustic noise, while suffering far less atmospheric attenuation than megahertz ultrasound used in biomedical scans.
2. **Q: Explain the mathematical derivation behind the divisor factor 58 in the distance formula.**
Ans: Speed of sound at 20°C is $v \approx 0.0343\text{ cm}/\mu\text{s}$. For two-way ToF: $\Delta t = \frac{2d}{v} = \frac{2 \times 1\text{ cm}}{0.0343\text{ cm}/\mu\text{s}} \approx 58.3\mu\text{s/cm}$.
3. **Q: Why is a $10\mu\text{s}$ pulse required on the Trigger pin?**
Ans: The $10\mu\text{s}$ active-HIGH pulse is the hardware start synchronization signal that prompts the internal controller of the HC-SR04 to emit its 8-cycle 40 kHz sonic burst.
4. **Q: What factors introduce measurement errors in ultrasonic ranging, and how can they be compensated?**
Ans: Errors arise from temperature variations ($v = 331.3 + 0.606 \times T$), soft sound-absorbing materials, angled targets causing specular deflection, and acoustic multipath reflections. Adding a temperature sensor (e.g., DHT11) provides dynamic speed compensation.
5. **Q: Why are separate transmitter (T) and receiver (R) transducer cans used in the HC-SR04 instead of a single transducer?**
Ans: Using dual discrete transducers isolates the sensitive receiver circuitry from the high-voltage excitation ringing of the transmitter, minimizing the minimum blind distance (2 cm) and simplifying the analog front-end design.
6. **Q: Why is a delay (60 – 100 ms) necessary between consecutive measurement cycles?**
Ans: Ultrasound travels at finite speed. A 100 ms delay ensures that residual echoes from the preceding cycle dissipate completely before launching a new pulse, preventing false triggers.